

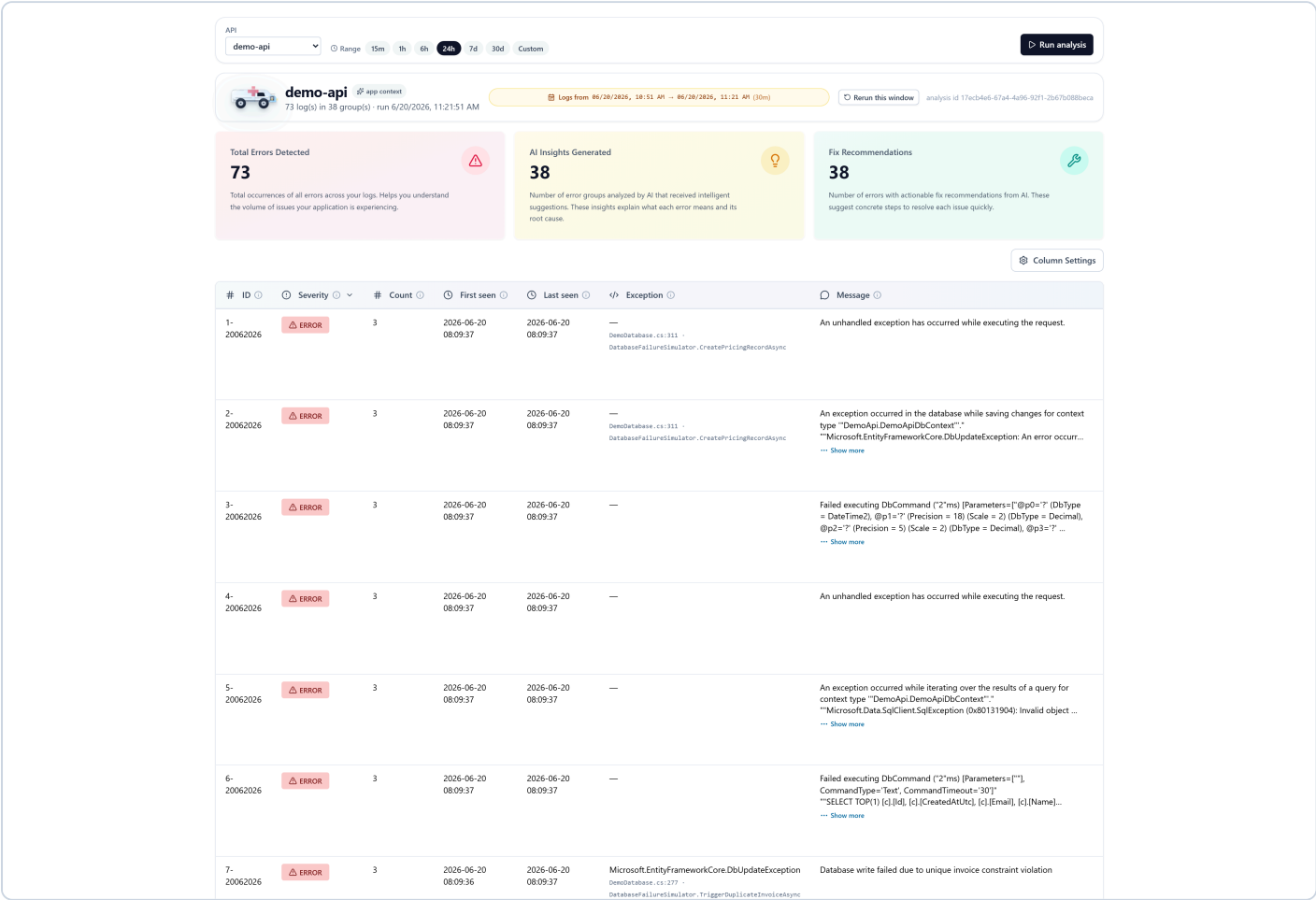
HotFixAmbulance — Working System Evidence

Production-error triage for .NET Web APIs, with AI analysis by a locally-hosted **Qwen** model. Captured 2026-06-20.

What it does & how it works. HotFixAmbulance pulls the last N hours of error logs from **Elasticsearch**, groups them by exception fingerprint, and for every group asks a locally-hosted **Qwen LLM** (**qwen2.5:3b**) — served in **Docker** through an Ollama-compatible runtime — to write a plain-English *“Suggestion for Error”* (what the error means) and a concrete *“How to fix”* (the remediation), grounded in the service’s Git history. Each run is persisted to **SQLite** and rendered in a **React** UI. Every group the model analyses is tagged analyzedBy = "Llm", and the UI renders a violet  badge on its two AI columns — so the use of Qwen is visible and verifiable, per group.

Pipeline: Elasticsearch → group by fingerprint → per-group Qwen enrichment (with Git-history grounding) → SQLite → REST API (:5283) → React UI (:5173).
Runtime: Qwen in Docker (CPU-only, :11434).

Figure 1 — System overview (ingestion, grouping, AI layer ran)



Triage run for the **demo-api** service: **73 error logs** collapsed into **38 distinct groups**. The metrics panel confirms the AI layer ran end-to-end — **38 “AI Insights Generated”** and **38 “Fix Recommendations”**, i.e. *every* group was enriched. The table shows each group’s severity, occurrence count, first/last-seen, exception type with stack frame, message and endpoint — the raw evidence read from Elasticsearch. **Proves:** the ingestion → grouping → analysis pipeline works end-to-end.

Figure 2 — Proof the analysis is produced by Qwen

API

demo-api

Range15m1h6h24h7d30dCustom

Run analysis

demo-apiapp context

73 log(s) in 38 group(s) · run 6/20/2026, 11:21:51 AM

Logs from 06/20/2026, 10:51 AM -- 06/20/2026, 11:21 AM (30m)

Rerun this window

analysis id 17ecb4e6-67a4-4a96-92f1-2b67b088beca

Total Errors Detected
73

Total occurrences of all errors across your logs. Helps you understand the volume of issues your application is experiencing.

AI Insights Generated
38

Number of error groups analyzed by AI that received intelligent suggestions. These insights explain what each error means and its root cause.

Fix Recommendations
38

Number of errors with actionable fix recommendations from AI. These suggest concrete steps to resolve each issue quickly.

Column Settings

#	ID	Severity	#	Count	Exception	Endpoint	Suggestion for Error	How to fix
1-	20062026	ERROR	3	—	DemoDatabase.cs:311 · DatabaseFailureSimulator.CreatePricingRecordAsync	/pricing/preview	Qwen The error likely stems from a failed database save operation at line 311 in DemoDatabase.cs, possibly due to a concurrency issue or an unhandled exception... Show more	Qwen Ensure that the DbContext is disposed before calling SaveChangesAsync to avoid InvalidOperationExceptions. Show more
2-	20062026	ERROR	3	—	DemoDatabase.cs:311 · DatabaseFailureSimulator.CreatePricingRecordAsync	/pricing/preview	Qwen The error occurs due to an invalid SQL object name 'PricingRecords' in the database, likely caused by a typo or schema change not reflected in the context. F... Show more	Qwen Update the DbSet for PricingRecords to match the actual table name in the database.
3-	20062026	ERROR	3	—	—	/pricing/preview	Qwen The error occurs when the database command fails, likely due to a parameter mismatch or missing value in the SQL query. The recent changes might have ... Show more	Qwen Review and update the parameters in the SQL query to match the expected types in the database. Check for any typos or missing values. Show more
4-	20062026	ERROR	3	—	—	/transfers/on-hold	Qwen The error is likely due to a refactor that introduced issues with the AsyncEnumerator.InitializeReaderAsync method in DefaultRules.cs. This could be caused by ... Show more	Qwen Review and revert or fix the AsyncEnumerator.InitializeReaderAsync method in DefaultRules.cs, specifically focusing on the changes introduced during the refactor.
5-	20062026	ERROR	3	—	—	/transfers/on-hold	Qwen The error occurs during a query execution, specifically when trying to access the 'Customers' table in the database. This is likely due to an invalid object name ... Show more	Qwen Update the entity framework model to reflect any changes made to the database. Specifically, ensure the 'Customers' table is correctly mapped. Show more
6-	20062026	ERROR	3	—	—	/transfers/on-hold	Qwen The error indicates a failed database query for a specific email, likely due to the customer not existing in the system. This issue may have been introduced b... Show more	Qwen Review and modify the query to handle cases where the customer does not exist. Specifically, add a check for the customer's existence before attempting the query. Show more
7-	20062026	ERROR	3	—	Microsoft.EntityFrameworkCore.DbUpdateException · DemoDatabase.cs:277 · DatabaseFailureSimulator.TriggerDuplicateInvoiceAsync	/invoices/duplicate	Qwen The error occurs because a unique constraint violation	Qwen Modify line 277 in DemoDatabase.cs to ensure that the unique constraint is properly handled before attempting to insert or update the invoice.

The two right-most columns are the AI columns. Every cell carries a violet  **Qwen** badge, which the UI renders *only* when a group's analyzedBy === 'LLM' — a value the backend sets *exclusively* after a successful Qwen completion (any model failure silently falls back to the heuristic and shows no badge). The text is genuinely model-generated and context-specific — e.g. “*The error likely stems from a failed database save operation at line 311 in DemoDatabase.cs...*” and “*The error occurs due to an invalid SQL object name 'Pricing.Records'...*” — not a static template. **Proves:** the system demonstrably uses Qwen to analyse each error and propose a fix.

Figure 3 — The complete result set (all 38 groups)

API

demo-api

Range15m1h6h24h7d30dCustom

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demo-apiapp context

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Logs from 06/20/2026, 10:51 AM -- 06/20/2026, 11:21 AM (30m)

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analysis id 17ecb4e6-67a4-4a96-92f1-2b67b088beca

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Number of errors with actionable fix recommendations from AI. These suggest concrete steps to resolve each issue quickly.

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3-	20062026	ERROR	3	—	—	—	/pricing/preview	Qwen The error occurs when the database command fails, likely due to a parameter mismatch or missing value in the SQL query. The recent changes might have ... Show more	Qwen Review and update the parameters in the SQL query to match the expected types in the database. Show more
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5-	20062026	ERROR	3	—	—	—	/transfers/on-hold	Qwen The error occurs during a query execution, specifically when trying to access the 'Customers' table in the database. This is likely due to an invalid object name ... Show more	Qwen Update the entity framework query to correctly reference the 'Customers' table. Show more
6-	20062026	ERROR	3	—	—	—	/transfers/on-hold	Qwen The error indicates a failed database query for a specific email, likely due to the customer not existing in the system. This issue may have been introduced b... Show more	Qwen Review and modify the query to ensure it correctly handles missing customer data. Show more
7-	20062026	ERROR	3	—	—	Microsoft.EntityFrameworkCore.DbUpdateException · DemoDatabase.cs:277 · DatabaseFailureSimulator.TriggerDuplicateInvoiceAsync	/invoices/duplicate	Qwen The error occurs because a unique constraint violation is being thrown when trying to save changes to the database, likely due to an existing invoice with the ... Show more	Qwen Modify line 277 in DemoDatabase.cs to ensure unique constraints are properly handled. Show more
8-	20062026	ERROR	3	—	—	DemoDatabase.cs:277 · DatabaseFailureSimulator.TriggerDuplicateInvoiceAsync	/invoices/duplicate	Qwen The error occurs because a duplicate invoice number is being inserted into the 'dbo.Invoices' table, violating its unique constraint. This likely happened due to ... Show more	Qwen Modify the code at demoDatabase.cs:277 to ensure unique constraints are properly handled. Show more
9-	20062026	ERROR	3	—	—	Microsoft.Data.SqlClient.SqlException	/payments/ab/settlement	Qwen The error occurs when the system fails to find a payment record by ID 'ab'. This likely stems from changes in the backend code, specifically around Enti... Show more	Qwen Review and update the Entity Framework query to ensure it correctly references the 'Payments' table. Show more
10-	20062026	ERROR	3	—	—	—	/payments/ab/settlement	Qwen The error occurs during a query execution, specifically when trying to read from the 'Payments' table. This likely stems from changes made in the ... Show more	Qwen Review and update the query to ensure it correctly references the 'Payments' table. Show more
11-	20062026	ERROR	3	—	—	—	/payments/ab/settlement	Qwen The error indicates a failed database query, likely due to the PaymentId parameter not matching any records in the Payments table. This issue may have been ... Show more	Qwen Review and modify the query to ensure it correctly validates the PaymentId parameter. Show more
12-	20062026	ERROR	3	—	—	Microsoft.Data.SqlClient.SqlException	/payments/ab	Qwen The error occurs when trying to fetch a payment by ID 'ab', likely due to the recent refactoring in Phase 10.1, which may have introduced issues. Show more	Qwen Review and update the query to ensure it correctly references the 'Payments' table. Show more
13-	20062026	ERROR	3	—	—	—	/payments/ab	Qwen The error indicates an invalid object name 'Payments' in the database query for context type 'DemoApi.DemoApiDbContext'. This likely stems from... Show more	Qwen Update the entity configuration to ensure the 'Payments' table name is correctly referenced. Show more

14-20062026	 ERROR	3	—	/payments/ab	 Qwen The error occurs when the API fails to execute a SQL query for retrieving a payment by its ID. This likely stems from changes made in the refactoring of the Show more	 Qwen Review and update the SQL query in the backend/src/HotfixAmbulance.cs specifically around the Show more
15-20062026	 ERROR	3	Microsoft.Data.SqlClient.SqlException	/orders	 Qwen The error indicates an invalid object name 'Customers' in the SQL query, likely caused by a change in database schema or a typo. The git history points to Show more	 Qwen Review and update the SQL query in the backend/src/HotfixAmbulance.cs to reflect the current database schema Show more
16-20062026	 ERROR	3	—	/orders	 Qwen The error occurs during a database query for the 'Customers' table, indicating an invalid object name. This likely stems from changes made in the Show more	 Qwen Review and modify the SQL query in the backend/src/HotfixAmbulance.cs specifically around the Show more
17-20062026	 ERROR	3	—	/orders	 Qwen The error occurs when trying to select a customer by their default email, likely due to the SQL query failing because there is no record matching the provided Show more	 Qwen Review and modify the SQL query in the backend/src/HotfixAmbulance.cs specifically around the Show more
18-20062026	 ERROR	2	—	/payments/ab/settlement	 Qwen The error indicates a 502 Bad Gateway HTTP response for the '/payments/ab/settlement' endpoint, likely caused by issues in the backend logic. This issue may Show more	 Qwen Review and modify the backend logic in the backend/src/HotfixAmbulance.cs specifically around the Show more
19-20062026	 ERROR	1	DenoDatabase.cs:311 - DatabaseFailureSimulator.CreatePricingRecordAsync	/pricing/preview	 Qwen The error indicates a 500 Internal Server Error on the '/pricing/preview' endpoint, likely caused by an issue with saving changes to the database in Show more	 Qwen Ensure that the call to _db.SaveChangesAsync(ct) is placed and does not cause Show more
20-20062026	 ERROR	1	—	/transfers/on-hold	 Qwen The error indicates a server-side HTTP 500 Internal Server Error on the '/transfers/on-hold' endpoint, likely caused by issues in the Show more	 Qwen Review and modify the ... method in the backend/src/HotfixAmbulance.cs Show more
21-20062026	 ERROR	1	—	/invoices/duplicate	 Qwen The API endpoint '/invoices/duplicate' responded with a 500 Internal Server Error, likely due to issues in the backend's SerilogDocumentMapper.cs file introduced... ... Show more	 Qwen Review changes made in the backend/src/HotfixAmbulance.cs focusing on the likely Show more
22-20062026	 ERROR	1	—	/users/-3	 Qwen The error likely stems from attempting to fetch a user with an invalid ID (-3). This could be due to improper validation or mapping of the input parameters in the Show more	 Qwen Review and modify the ... method in the backend/src/HotfixAmbulance.cs specifically around the Show more
23-20062026	 ERROR	1	System.ArgumentOutOfRangeException	/users/-3	 Qwen The error occurs when trying to look up a user with an invalid ID (-3). This likely stems from the recent refactoring in DefaultRules.cs, where Stanislav-Stanev... ... Show more	 Qwen Review and modify the ... method in the backend/src/HotfixAmbulance.cs to ensure only valid user Show more
24-20062026	 ERROR	1	—	/orders	 Qwen The error indicates a server-side HTTP 500 Internal Server Error when handling POST requests to the '/orders' endpoint. This likely stems from issues in the... ... Show more	 Qwen Review and modify the ... method in the backend/src/HotfixAmbulance.cs (2026-06-18) by Stanislav-Stanev Show more
25-20062026	 ERROR	1	DenoDatabase.cs:311 - DatabaseFailureSimulator.CreatePricingRecordAsync	/pricing/preview	 Qwen The error occurs because the database save operation is not awaited, leading to an incomplete transaction and a potential data inconsistency or loss.	 Qwen Await the call to _db.SaveChangesAsync(ct) in the backend/src/HotfixAmbulance.cs Show more

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The full run. **All 38 groups** carry the  badge on both AI columns, confirming the entire run was analysed by Qwen (run-level analyzedBy = "L1m"), not the heuristic fallback. **Proves:** the whole demo flow runs on Qwen.

Figure 4 — Backend / runtime evidence (terminal)

```
# The Qwen model is served from Docker (CPU-only) on :11434
$ docker ps --filter name=hfa-qwen
hfa-qwen    ollama/ollama:latest    Up 59 minutes (healthy)    0.0.0.0:11434->11434/tcp

# The qwen2.5:3b model is pulled into the persistent volume
$ docker exec hfa-qwen ollama list
NAME        ID                SIZE    MODIFIED
qwen2.5:3b  357c53fb659c      1.9 GB  37 minutes ago
```

```
# A triage run produced via the API – analyzed by the LLM, every group enriched
$ curl -s http://localhost:5283/api/triage/runs/17ecb4e6-.../
apiName      = demo-api
analyzedBy   = Llm
totalLogs    = 73
totalGroups  = 38
summary.withSuggestions = 38      # all 38 groups got an AI suggestion
summary.withFixes      = 38      # all 38 groups got an AI fix

# During inference the Qwen container saturates the CPU (one call per group)
$ docker stats hfa-qwen --no-stream
hfa-qwen   CPU = 1177%   MEM = 4.4 GiB / 15.4 GiB
```

Proves: Qwen runs locally in Docker, the model is loaded, and a real triage run returns analyzedBy = Llm with an AI suggestion + fix for every one of the 38 groups.

How the project is built (modules)

- **Ingestion** — HotFixAmbulance.Elastic: queries Elasticsearch for Fatal/Error/Warning logs in the window.
- **Analysis / grouping** — HotFixAmbulance.Analysis: groups logs by (exception, normalized message, endpoint) and ranks them.
- **AI enrichment** — LlmGroupEnricher + OllamaLlmClient: calls the Qwen runtime per group; on any failure falls back to a deterministic Git-history heuristic.
- **Git insights** — HotFixAmbulance.GitInsights: mines origin/main for blame + related commits to ground the model.
- **Persistence** — HotFixAmbulance.Persistence: stores each run (and per-group AnalyzedBy) in SQLite.
- **API + UI** — ASP.NET Core REST API and a React/TanStack table that renders the 🚑 Qwen badge.
- **Infrastructure** — infra/qwen: Docker Compose Qwen runtime + bootstrap that pre-pulls the model (with corporate-CA injection for TLS-intercepting proxies).

Repository: <https://github.com/myPOStech/mps-banking-hot-fix-ambulance> · **AI tool used:** Claude Code (Opus/Sonnet) for design, implementation, TDD and orchestration.